

Kavya Murali

Full Stack Software Engineer | AI Engineer

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🌐 Portfolio

Professional Summary

Software Engineer with 3+ years of experience building reliable, performant backend systems in **C#/.NET, React, Next, Python, TypeScript and SQL**, including **AI-powered and LLM-integrated features**. At Zoho, develops desktop applications for **Zoho Mail, Notes and Contacts** used by **10,000+** enterprise customers: REST API and **SMTP/IMAP/POP** integrations, SQLite persistence and sync, full-text search, AI feature integration across application, API and data layers, and production incident resolution via root-cause analysis and profiling. Designed an end-to-end **RAG pipeline** (FastAPI, Qdrant, embeddings, reranking) that raised retrieval accuracy from **61% to 88%** through evaluation-driven tuning. Independently architected multi-tenant data-access layers, RBAC, fail-closed authorization middleware and additive-only migration strategies across PostgreSQL and MongoDB. Grounded in SOLID, clean architecture, async programming and regression testing.

Technical Skills

Languages: C#, Python, TypeScript, JavaScript (ES6+), SQL, Java, C, HTML5, CSS3, XAML

AI & Retrieval: RAG Pipelines, LLM Integration, Prompt Engineering, Embeddings, Semantic & Hybrid Search, Reranking, Chunking Strategies, Retrieval Evaluation (RAGAS), Vector Databases (Qdrant, Pinecone), Hugging Face Transformers, WebLLM, Local/Browser Inference

Backend & Frameworks: .NET, ASP.NET, FastAPI, Node.js, Express.js, Next.js (App Router, Server Actions), NextAuth, REST API Design, Zod Validation, Middleware, Cron Jobs

Frontend: React 19, Redux, Vite, Tailwind CSS, Responsive UI, Component Architecture, UWP/XAML, MVVM

Databases: PostgreSQL (pg_trgm, GIN indexes, query planning), MongoDB (compound & partial indexes), SQLite, Supabase, Prisma ORM, Qdrant, Schema Design, Migrations, Pagination, Query Optimization

Security & Multi-tenancy: Authentication & Authorization, JWT, RBAC, Session Management, bcrypt, Tenant-scoped Data Access, Audit Logging, Fail-closed Middleware

Cloud & DevOps: Vercel (Deployments, Cron), AWS S3, Supabase CLI, Git, GitHub, CI/CD, Environment-gated Deployments

Integration & Protocols: REST, JSON, XML, SMTP, IMAP, POP, Postman, PDF/Excel Report Generation, SEO-friendly Routing

Engineering Practices: SOLID, OOP/OOD, Design Patterns, Clean Architecture, Multithreading & Async Programming, Full-Text Search, Unit & Regression Testing, Code Review, Root-cause Analysis, Performance Profiling, SDLC

Work Experience

Member Technical Staff (Software Engineer) – Zoho Corporation

06/2023 – Present | Chennai, India

- Developed and maintained **C#/.NET UWP desktop applications** for **Zoho Mail, Zoho Notes and Zoho Contacts**, shipping production features to **10,000+** enterprise customers.
- Built and integrated **REST APIs** and email workflows over **SMTP, IMAP and POP**, covering synchronization, send/receive and account-level operations.
- Implemented **SQLite persistence** and sync with optimized data-access paths, keeping the UI responsive on large local datasets; tuned application, database, sync and API performance by removing bottlenecks and redundant processing.
- Delivered AI-powered user-facing features integrated across the application, API and data layers of existing product workflows.
- Resolved **production incidents** via root-cause analysis, log investigation and performance profiling, reducing recurring failures and improving reliability.
- Applied SOLID, MVVM, async programming and reusable component design; maintained quality through **unit tests, peer code reviews** and cross-team collaboration.

React Developer (Part-time) – Techdome Solutions

06/2021 – 06/2022 | Remote

- Built **20+** reusable React/Redux components across **8** screens of a responsive web app, integrating **15+** REST endpoints and cutting duplicated UI code by **~30%**.
- Authored **100+** unit tests and reviewed **8+** pull requests weekly in a team of **6** engineers, improving coverage and release stability.

Projects

Personal Memory Engine

2026

Python, FastAPI, React (Vite), Qdrant, Hugging Face, WebLLM

- Built a personal knowledge-retrieval system indexing **1,200+** documents from WhatsApp, Instagram and journal exports into a **Qdrant** vector store serving semantic search over **18,000+** embeddings.
- Implemented an end-to-end **RAG pipeline** (chunking, embedding, retrieval, reranking, response synthesis) and raised retrieval accuracy from **61% to 88%** on a 25-question evaluation set via chunk-size and overlap tuning.
- Designed a **privacy-first local/cloud mode**: with Privacy Mode on, filtering and inference run in-browser using **Hugging Face Transformers and WebLLM** with browser-local storage.
- Added semantic photo retrieval for **500+** uploaded photos with metadata-aware filtering and vector search, cross-linking images to related chats and journal entries.

Client Task Tracker (CGrid)

2026

Next.js 16, TypeScript, Supabase, PostgreSQL, NextAuth

- Architected a **multi-tenant data-access layer** centralizing DB access through a reusable `DBUtil` abstraction with 4 tenant-scoping primitives, enforcing isolation and authorization boundaries across 8 tables without Postgres RLS.
- Implemented authentication and **RBAC with NextAuth**: session management, fail-closed middleware, 401/403 gates, forced password-reset flow, **bcrypt** hashing and audit logging of role changes and mutations.
- Improved **PostgreSQL** search latency with **6 pg_trgm GIN indexes**; bounded latency, memory and payload size with pagination (100 records/page).
- Managed schema evolution through **15 additive-only SQL migrations** via the **Supabase CLI** (multi-tenancy, status pipelines, audit logging) with environment-gated CI/CD deployments on Vercel.
- Established **Zod schema validation** and endpoint-level regression checklists covering auth, RBAC, payload validation and API contracts across customer, task and dues APIs.

Corazon – Team Time Tracking & Billing Platform

2026

Next.js 16, React 19, MongoDB, Vercel

- Architected a **multi-tenant SaaS platform** with 11+ data models and organization-scoped isolation; compound indexing cut query latency from **2.3s to 180ms** while supporting **1,000+** users and 10,000+ time logs.
- Engineered security with **JWT, AsyncLocalStorage request context, middleware authorization and 4-role RBAC** (Super-Admin, Admin, Lead, Employee), enforcing row-level access to prevent cross-tenant reads.
- Built a time-tracking and attendance pipeline with timer state management, server-side validation, audit trails, partial unique indexes and **Vercel Cron** midnight (IST) closeout for accurate billing hours.
- Delivered reporting and analytics (client/date filters, rate-based billing, dashboard charts, automated **PDF and Excel export**), reducing manual reporting from **30 minutes to one click**; shipped **50+ feature releases** within free-tier infrastructure limits.

Algae Research Database & Portfolio Site

2025

Next.js, React, TypeScript, Prisma, SQLite, AWS S3

- Built and deployed a full-stack research database cataloging **258 species** across 9 divisions, 14 classes and 69 families, with dynamic category routing and **250+ SEO-optimized species pages**.
- Designed a normalized **Prisma/SQLite schema** (5 tables) covering taxonomy, synonyms, ecology and distribution; managed **13 Prisma migrations** for version-controlled schema evolution.
- Published **58 academic resources** and a searchable archive of **24 historical publications** via a Student Corner portal; integrated **AWS S3** for media delivery across all species records.

Education

Bachelor of Engineering, Electrical and Electronics Engineering – St. Joseph's College of Engineering, Chennai

2019 – 2023

Aggregate: 94%